Lecture 5 - July 13th (Antiderivatives)

Math 225 - Summer.

Function $\xrightarrow{\text{Math 224}}$ Derivative.

$f(x)$ $\xleftarrow{\text{Math 225.}}$ $f'(x)$

Math 225 if you have a function $g(x)$ is this function the result of compute the derivative of some other function?

For example if $g(x) = 2x$ then $f_1(x) = x^2$ is a function s.t $f_1'(x) = 2x = g(x)$. $f_2(x) = x^2 + 1$, $f_3(x) = x^2 - 5$.

Def A function F is called an antiderivative of f on an interval I if $F'(x) = f(x)$ for all $x \in I$.

Thm: If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x) + C.$$

where C is an arbitrary constant.

Corollary If F and G are antiderivatives of f then

$$F - G = C.$$

Examples: Find the most general antiderivative of

i). $f(x) = \sin x$

ii). $f(x) = x^n$, $n \geq 0$

iii). $f(x) = x^{-3}$

Sln: i). $G(x) = -\cos x$ b/c $G'(x) = \frac{d}{dx}(-\cos x) = -\frac{d}{dx}(\cos x) = -(-\sin x) = \sin x$.

So, the most general antiderivative is $F(x) = -\cos x + C$.

ii). $G(x) = \frac{x^{n+1}}{n+1} + C$. then $G'(x) = \frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) + \frac{d}{dx}(C)$

$$= \frac{1}{n+1} \cdot \frac{d}{dx}(x^{n+1}) = \frac{1}{n+1}(n+1)x^{(n+1)-1}.$$

$= x^n$.

iii). $H(x) = \dfrac{x^{-3+1}}{-3+1} + c = \dfrac{x^{-2}}{-2}$ then $H'(x) = \dfrac{-2}{-2} x^{-2-1} = x^{-3}$.

Table of antiderivatives formulas:

Function	Particular Antiderivative
$\cos x$	$\sin x$
$\sin x$	$-\cos x$
$\sec^2 x$	$\tan x$.
$\sec x \tan x$	$\sec x$
x^n, $n \neq 1$	$\dfrac{x^{n+1}}{n+1}$
$\dfrac{1}{2\sqrt{x}}$	$\sqrt{x}$

Example. Find all functions g such that.

$$g'(x) = 4\sin x + \frac{2x^5 - \sqrt{x}}{x}.$$

Sln: $g'(x) = 4\sin x + \dfrac{2x^5}{x} - \dfrac{\sqrt{x}}{x}$; $\dfrac{x^n}{x^m} = x^{n-m}$.

$$g'(x) = 4\sin x + 2x^4 - x^{-1/2}.$$

Thus, we want to find an antiderivative of $4\sin x + 2x^4 - x^{-1/2}$.

Antiderivative of: $4\sin x$ is $-4\cos x$.

Antiderivative of: $2x^4$ is $2 \cdot \dfrac{x^5}{5}$

Antiderivative of: $x^{-1/2}$ is $\dfrac{x^{-1/2+1}}{-1/2+1} = \dfrac{x^{1/2}}{1/2} = 2\sqrt{x}$.

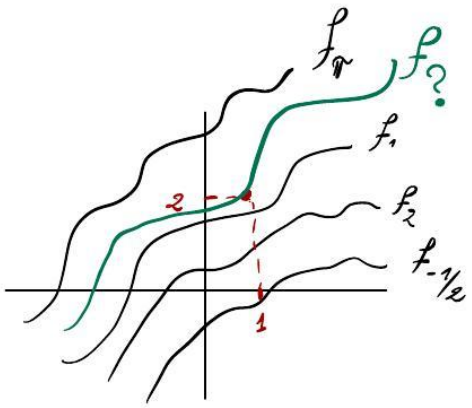
then

$$g(x) = -4\cos x + \frac{2}{5}x^5 - 2\sqrt{x} + C.$$

Example: Find f is $f'(x) = x\sqrt{x}$ and $f(1) = 2$.

Sln: $f'(x) = x \cdot x^{1/2} = x^{3/2}$ So

$$f(x) = \frac{x^{3/2+1}}{3/2+1} + C = \frac{2}{5}x^{5/2} + C.$$



then $f(1)=2=\frac{2}{5}(1)^{5/2}+C$.

$$C=2-\frac{2}{5}=\frac{8}{5} \quad \text{then}$$

$$f(x)=\frac{2}{5}x^{5/2}+\frac{8}{5}$$

Example. Find f if $f''(x)=12x^2+6x-4$; $f'(0)=4$, $f(1)=1$.

Sln: The general derivative of $f''(x)$ is

$$f'(x)=12\cdot\frac{x^{2+1}}{2+1}+6\cdot\frac{x^{1+1}}{1+1}-4\frac{x^{0+1}}{0+1}+C$$

$$f'(x)=\frac{12}{3}x^3+\frac{6}{2}x^2-4x+C$$

$$f'(x)=4x^3+3x^2-4x+C.$$

Using $f'(0)=4$ then

$$f'(0)=C=4$$

$$f'(x)=4x^3+3x^2-4x+4$$

then

$$f(x)=\frac{4x^4}{4}+\frac{3x^3}{3}-\frac{4x^2}{2}+4x+D$$

$$f(x)=x^4+x^3-2x^2+4x+D.$$

then $f(1)=1$

$$1=f(1)=1^4+1^3-2(1)^2+4(1)+D$$

$$1=2-2+4+D$$

$$D=-3$$

So,

$$f(x)=x^4+x^3-2x^2+4x-3$$

Linear Motion:

Example: A particle moves in a straight line and has acceleration given by $a(t)=6t+4$. Its initial velocity is $v(0)=-6$ cm/s

and its initial displacement is $s(0) = 9$ cm. Find its position function.

Sln: $s(t) \Rightarrow s'(t) = v(t) \Rightarrow s''(t) = v'(t) = a(t)$.

Since $v'(t) = a(t) = 6t + 4$ then

$$v(t) = \frac{6t^2}{2} + 4t + C.$$

$$v(t) = 3t^2 + 4t + C.$$

$$-6 = v(0) = 0 + 0 + C \Rightarrow C = -6.$$

$$\boxed{v(t) = 3t^2 + 4t - 6.}$$

Since $s'(t) = v(t)$ then, the antiderivative of $v(t)$ is

$$s(t) = t^3 + 2t^2 - 6t + D.$$

$$9 = s(0) = 0 + 0 + 0 + D \Rightarrow D = 9.$$

$$\boxed{s(t) = t^3 + 2t^2 - 6t + 9}$$

Example: A ball is thrown upward with a speed of 48 ft/s from the edge of a cliff, 432 ft above the ground. Find its height above the ground t second later. When does it reach its maximum height? When does it hit the ground?

problem for you!

$a(t) = -32$.

(Appendix E: Sigma Notation and 4.1 Areas and Distances)

Def. If $a_m, a_{m+1}, _, a_n$ are real numbers and m and n are integers numbers such that $m \le n$ then

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + a_{m+2} + \cdots + a_n.$$

Notation: this tell us to end with $i=n$; this tell us to add a_i; this tell us to start with $i=m$

$$\sum_{i=m}^{n} a_i$$

Examples: i). $\sum_{i=1}^{4} i^2 = 1^2+2^2+3^2+4^2$

ii). $\sum_{i=3}^{n} i = 3+4+5+\cdots+(n-1)+n$

iii). $\sum_{j=0}^{5} 2^j = 2^0+2^1+2^2+2^3+2^4+2^5 = 63.$

iv) $\sum_{k=1}^{n} \frac{1}{k} = 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}.$

v). $\sum_{i=1}^{3} \frac{i-1}{i^2+3} = \frac{1-1}{1^2+3} + \frac{2-1}{2^2+3} + \frac{3-1}{3^2+3} = \frac{13}{42}.$

Example: write the sum $2^3+3^3+\cdots+n^3$ in sigma notation.

Sln: There is no unique way of writing a sum in sigma notation

$$2^3+3^3+\cdots+n^3 = \sum_{i=2}^{n} i^3$$

$$= \sum_{j=1}^{n} (j+1)^3$$

$$= \sum_{k=0}^{n} (k+2)^3.$$

Thm. If $c \in \mathbb{R}$ is any constant, then

i). $\sum_{i=1}^{n} c \cdot a_i = c \cdot \sum_{n=1}^{n} a_i$ ii). $\sum_{i=1}^{n} (a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$

Rmk: find $\sum_{i=1}^{n} 1 = \underbrace{1+1+1+\cdots 1}_{n\text{-terms}} = n.$

Thm: Let c be a constant and n a positive integer. Then.

a). $\sum_{i=1}^{n} 1 = n.$

b). $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ if $\sum_{i=1}^{5} i = \frac{5(5+1)}{2} = \frac{5(6)}{2} = 15$

c). $\sum_{i=1}^{n} c = n \cdot c$

d). $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

e). $\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$

Example. Evaluate $\sum_{i=1}^{n} i(4i^2 - 3i)$.

Sln:

$$\sum_{i=1}^{n} i(4i^2 - 3i) = \sum_{i=1}^{n} (4i^3 - 3i^2)$$

$$= \sum_{i=1}^{n} 4i^3 - \sum_{i=1}^{n} 3i^2$$

$$= 4 \cdot \sum_{i=1}^{n} i^3 - 3 \sum_{i=1}^{n} i^2$$

$$= 4\left\{\left(\frac{n(n+1)}{2}\right)^2\right\} - 3\left\{\frac{n(n+1)(2n+1)}{6}\right\}$$

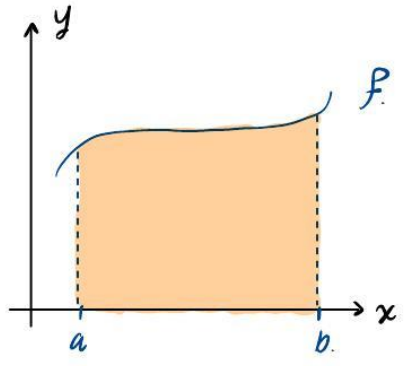
$$= n^2(n+1)^2 - \frac{1}{2} n(n+1)(2n+1)$$

$$= n(n+1)\left\{n(n+1) - \frac{2n+1}{2}\right\}$$

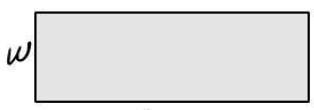
$$= n(n+1)\left(\frac{2n^2+2n-2n-1}{2}\right)$$

$$= \frac{n(n+1)(2n^2-1)}{2}.$$

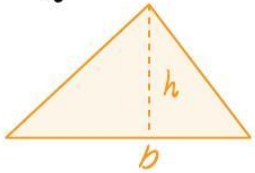
The Area Problem. We know from Math 224 that if $y = f(x)$ the the graphic is.



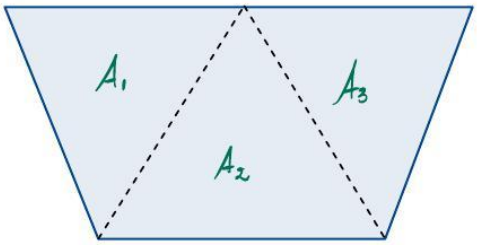
We know that



$A = l \cdot w$

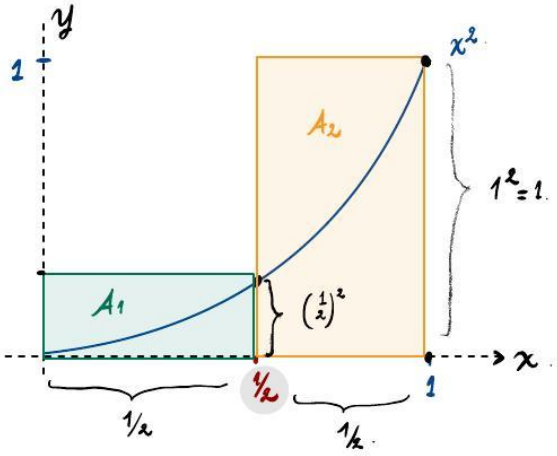


$A = \frac{1}{2} b \cdot h$

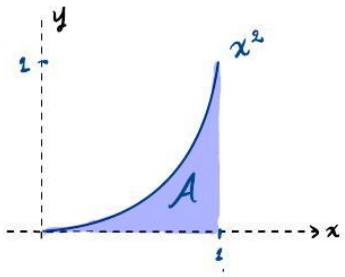


$A = A_1 + A_2 + A_3$.

Example: Use rectangles to estimate the area under the parabola $y = x^2$ from $x = 0$ to $x = 1$.



$A \approx A_1 + A_2$



$$A = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)(1)^2 = \frac{1}{8} + \frac{1}{2} = \frac{1+4}{8} = \frac{5}{8}.$$

For you! to approximate the area using three rectangles.